

RESEARCH PROPOSAL

Alcohol Access Laws & Road Safety Externalities

A Spatial Difference-in-Differences Analysis of County Wet/Dry Status Changes

Behavioral Economics · Crime & Antisocial Behavior · Transportation Policy

~500

U.S. Counties

remain fully or partially dry

13,500+

Alcohol-Related Fatalities/yr

NHTSA annual average

15+

States with Active Debates

Southern & Midwestern states

Dry counties may inadvertently increase drunk driving by forcing residents to travel farther for alcohol — but existing evidence relies on outdated methods and ignores spatial spillovers.

Literature Gap Prior wet/dry work uses TWFE on short panels without spatial controls. No existing study applies heterogeneity-robust DiD or explicitly models cross-border spillovers to neighboring counties.

RESEARCH QUESTION

When a dry county becomes wet, how do traffic accidents and vehicle miles traveled change — both within the newly-wet county and in neighboring dry counties that previously served as the closest alcohol access point?

- ▶ **Primary** Effect on fatal crashes & VMT in treated (newly-wet) counties
- ▶ **Secondary** Spillover effects on neighboring dry counties
- ▶ **Heterogeneity** On-premise vs. off-premise licensing; beer/wine vs. spirits
- ▶ **Marginal Contribution** First heterogeneity-robust DiD with explicit spatial spillover design in the wet/dry literature; VMT as behavioral mechanism test

Effective Price of Alcohol

$$U_i = u(a_i) + x_i \text{ s.t. } p \cdot a_i + \tau(d_i) \cdot a_i + x_i = y_i$$

Effective price: $p^* = p + \tau(d_i)$, $\tau'(d) > 0$.

Dry county: $d_i = \bar{d}_i > 0 \Rightarrow p^{*\text{dry}} > p^{*\text{wet}}$
 $\Rightarrow$ lower equilibrium consumption.

Impaired Miles Driven

$$m_i = \alpha a_i + \beta d_i$$

Accident risk: $\pi_i = \pi(a_i, m_i)$, $\partial\pi/\partial a > 0$,
 $\partial\pi/\partial m > 0$.

Going wet eliminates the βd_i term.

Key Comparative Static

$$\frac{d\pi_i}{dD_c} = \underbrace{\frac{\partial\pi}{\partial a} \frac{\partial a}{\partial p^*} \frac{\partial p^*}{\partial D_c}}_{\text{consumption (+)}} + \underbrace{\frac{\partial\pi}{\partial m} \beta \frac{\partial d_i}{\partial D_c}}_{\text{travel dist. (-)}}$$

Net effect is ambiguous — motivates empirical work.

Testable Predictions

1. Consumption $\uparrow$ in newly-wet counties
2. Net accident change: empirical question
3. Neighboring dry counties: fewer long-haul impaired trips
4. VMT for alcohol-related travel falls overall

Source	Full Name	Role & Notes	Sample / Grain	Type
FARS	<i>Fatality Analysis Reporting System</i>	Fatal crashes with alcohol flags. Crash-level records aggregated to county × year. Full extract 2000–2023; estimation uses 2008–2023 (pre-2008 quality insufficient).	AR, 75 counties; county × year	Primary Outcome
AR Wet/Dry	<i>Hand-compiled from legislative records & news</i>	Transition dates 2010–2022, verified via AR ABC maps and election coverage. Neighbor matrix from U.S. Census county adjacency file (2023).	AR, 12 transitions; county × year	Treatment Var.
IPUMS	<i>Multiple IPUMS surveys (ACS & other)</i>	Demographics, income, rurality, labor-market controls. Addresses population sorting and baseline heterogeneity.	AR counties; annual	Controls
HPMS[†]	<i>FHWA Hwy. Performance Monitoring Sys.</i>	VMT by county & road type. Behavioral mechanism test for the travel-distance channel.	AR counties; annual	Planned

[†] HPMS VMT not yet collected; all other datasets assembled and merged (975 county-year obs., 2008–2024). FARS fatalities preferred over injury crashes: lower misreporting risk.

METHODS — Staggered Difference-in-Differences with Spatial Spillovers

Dep. Var.: Fatal crashes & VMT per county-year **Key Indep. Var.:** Wet/dry status (binary) + neighbor treatment indicator **Key Variation:** Staggered timing of county legalization events

Panel Structure

County $\times$ year panel, Arkansas 2008–2023. Staggered treatment timing from 12 wet/dry transitions provides clean identifying variation without reliance on a single policy shock.

Spatial Spillover Design

Contiguous dry counties to a newly-wet county receive a *neighbor treatment* indicator, constructed from the U.S. Census Bureau county adjacency file. Distance bands (25 mi, 50 mi) used as robustness checks.

Modern DiD Estimator

TWFE is biased under treatment effect heterogeneity (Callaway & Sant’Anna 2021; Sun & Abraham 2021). We use **Callaway & Sant’Anna (2021)** group-time ATT estimator throughout, with event-study plots to verify pre-trends.

Fixed Effects

- ▶ **County FE:** time-invariant culture, geography, baseline drinking norms
- ▶ **Year FE:** national drunk-driving trends, vehicle safety improvements

Summary Statistics (2010–2020 Window)

	Mean per county-year
Fatal crashes	6.61
Alcohol-involved	1.83
Alcohol share	26.9%
Treated counties	12
Neighbor counties	63
Total county-years	825

Staggered Treatment Cohorts

County	Cohort	Pre / Post
Boone, Clark	2010	2yr / 10yr
Benton, Sharp	2012	4yr / 8yr
Saline, Columbia	2014	6yr / 6yr
Little River	2016	8yr / 4yr
Randolph	2018	10yr / 2yr
Van Buren, Sevier	2020	12yr / 0yr

Zero clean never-treated controls in Arkansas $\Rightarrow$ neighbor-as-control design required. Cohort variation in pre-period length (2–12 years) enables robust pre-trend testing across all groups.

Reporting Bias

Alcohol-involved crash rates vary by state/sheriff department.

Fix: Use fatalities (FARS) as primary outcome; harder to misreport.

Population Sorting

Going wet may attract different residents over time.

Fix: Focus on short-run effects; include IPUMS demographic controls; test with border-discontinuity sample.

Staggered DiD Bias

TWFE is biased when treatment effects are heterogeneous (Callaway & Sant'Anna 2021; Sun & Abraham 2021).

Fix: Callaway & Sant'Anna / Sun & Abraham estimators used throughout as primary specification.

Wet/dry laws remain actively debated in 15+ states. Clean causal evidence directly informs ongoing legislative discussions in Southern and Midwestern states.

01 Methodological

First application of Callaway & Sant'Anna / Sun & Abraham heterogeneity-robust estimators to the wet/dry literature, addressing a known gap in prior panel work.

02 Spatial Externality Framing

If spillover estimates show significant reductions in neighboring counties, this reframes dry laws as a *redistribution* of accident risk rather than a net reduction — with direct implications for regional policy design.

03 Behavioral Mechanism

VMT as an outcome alongside accidents provides evidence on the travel-distance channel, distinguishing consumption-driven from distance-driven accident risk.

04 Welfare Analysis

Net externality accounting for spatial redistribution: compute total social cost across treated + neighbor counties to assess whether legalization creates aggregate safety gains or merely relocates them.

Data Assembly

- ▶ Collect HPMS VMT county $\times$ year (primary gap)
- ▶ Finalize IPUMS variable selection across surveys
- ▶ Extend FARS estimation window to 2023

Empirical Setup

- ▶ Define neighbor treatment (adjacency file + distance bands)
- ▶ Construct staggered treatment indicators
- ▶ Implement C&S-A estimator; generate event-study plots
- ▶ Verify pre-trends across all cohorts

Extensions

- ▶ Heterogeneity by legalization type (on/off-premise)
- ▶ Separate beer/wine from spirits
- ▶ Welfare analysis: net externality accounting for spatial redistribution
- ▶ Border-discontinuity robustness sample

Open to advisor feedback on HPMS data prioritization and neighbor definition strategy.

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